Project Initialization and Planning Phase

Date	15 July 2026
Team ID	
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Project Overview

Category	**Description**
Objective	The primary objective is to develop a machine learning-based Credit Card Approval Prediction system that analyzes customer financial information and predicts approval eligibility accurately and efficiently. The system aims to reduce manual evaluation time, improve decision-making, and provide transparent approval predictions.
Scope	The project focuses on automating the credit card approval process by using machine learning algorithms to evaluate factors such as income, credit history, employment status, and personal details. The system provides quick predictions and helps customers understand their approval chances.

ProblemStatement

Category	Description
Description	Traditional credit card approval processes are time-consuming and depend heavily on manual verification, which can lead to delays, errors, and inconsistent decisions. Customers often lack clarity about their eligibility before applying.
Impact	Solving these issues improves operational efficiency, reduces approval risks, minimizes manual effort, and enhances customer satisfaction by providing faster and more accurate credit approval predictions.

Category	Description
Proposed Solution	

Category	Description
Approach	Implement a machine learning-based prediction system that analyzes applicant data and predicts whether a credit card application is likely to be approved. The model will learn from historical customer data to improve accuracy and support automated decision-making.
Key Features	• Machine learning-based credit approval prediction model • Real-time prediction of approval probability • Data preprocessing and feature analysis for accurate results • User-friendly interface for entering applicant details • Performance evaluation of ML models using accuracy metrics • Recommendations to improve approval chances

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	Processing resources required for model training and prediction	CPU/GPU support (T4 GPU recommended for faster ML processing)
Memory	RAM required for handling datasets and model training	8 GB RAM
Storage	Space required for datasets, trained models, and application files	1 TB SSD Storage
Software		
Frameworks	Backend development framework	Python Flask
Libraries	Machine learning and data processing libraries	Scikit-learn, Pandas, NumPy,

Category	**Description**	
		XGBoost, Joblib
Data Visualization Tools	Used for analysis and visualization	Matplotlib, Seaborn
Development Environment	Coding and model development tools	Jupyter Notebook, PyCharm
Data		
Dataset	Data used for training and testing the prediction model	Kaggle Credit Card Approval Dataset / UCI Credit Approval Dataset
Dataset Format	Structured data containing applicant attributes	CSV format (Kaggle: 614 records, UCI: 690 records)